

SOC - HOME LAB

NETWORK INFRASTRUCTURE & SECURITY LAB

A home lab environment built for learning, experimentation, and hands-on experience in cybersecurity, network monitoring, infrastructure management, and SOC analyst skills.



LEARN



BUILD



MONITOR



DETECT



RESPOND



IMPROVE

LAB PURPOSE

- ✓ Develop SOC Analyst & Network Engineering skills
- ✓ Build visibility with real-world telemetry
- ✓ Practice detection engineering & threat hunting
- ✓ Experiment with open-source & enterprise tools
- ✓ Document, learn, and continuously improve



SECURE
MONITOR
ANALYZE

DETECT THREATS.
PROTECT SYSTEMS.
STAY CURIOUS.

HARDWARE INVENTORY



ARRIS SURFboard SB8200 Cable Modem

- DOCSIS® 3.1
- Up to 2 Gbps Downstream
- Multi-Gigabit Capable



OpenWrt One Router (Gateway)

- ARMv8 / MediaTek
- OpenWrt 24.03 / Kernel 6.6.x
- Gigabit Ports
- DHCP / DHCPv6
- WiFi Access Point



Netgear GS108Tv3 Smart Managed Switch

- 8x Gigabit Ports
- VLAN Support
- Port Mirroring
- Traffic Management
- Cloud Management



Raspberry Pi 5 (8GB) Analytics Node

- Quad-Core Cortex-A76
- 8GB LPDDR4X RAM
- Gigabit Ethernet
- Docker Environment



AXIS M1075-L Network Camera

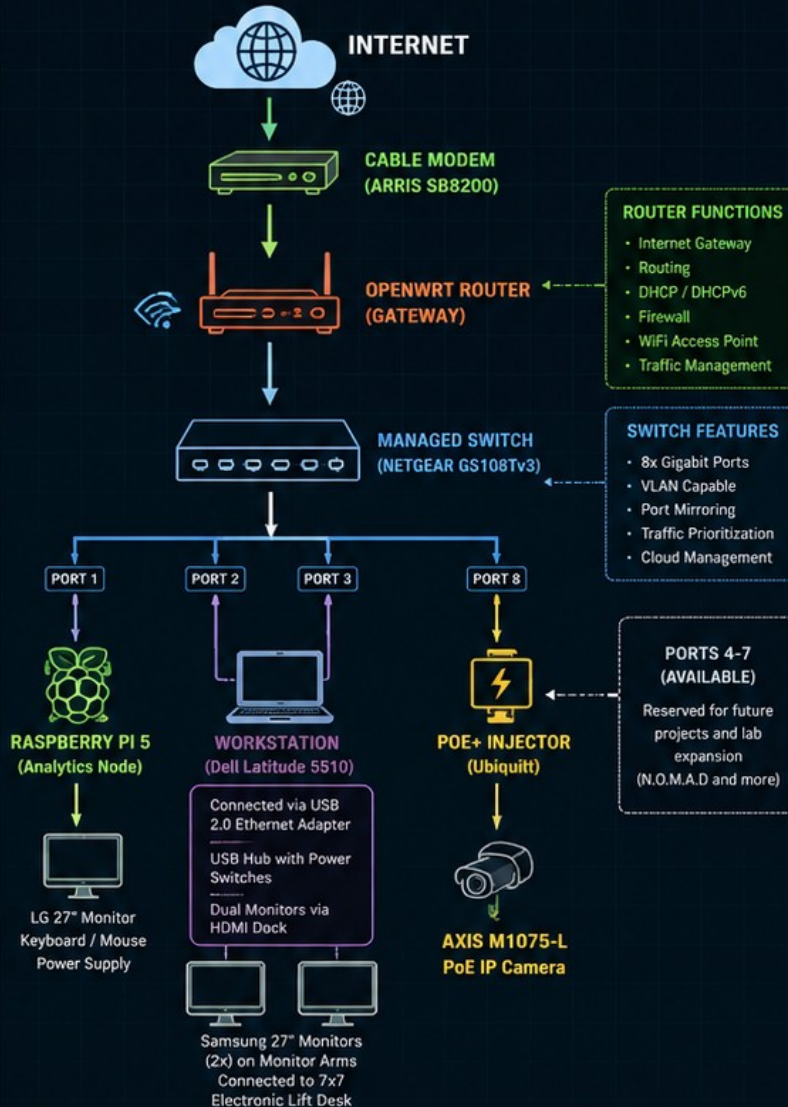
- PoE Powered
- RTSP Streaming
- microSD Local Storage



Dell Latitude 5510 Workstation

- Intel i5-10310U
- 8GB RAM | 238GB Storage
- Windows 11 Pro
- Primary Engineering System

NETWORK INFRASTRUCTURE TOPOLOGY



ADDITIONAL DEVICES (THROUGH GATEWAY)



Samsung Smart TV
(Living Room)



Vizio Smart TV
(Bedroom)



Alexa
(Living Room)



Echo Dot
(Master Bedroom)



Samsung Galaxy S24+
(Stacey's Phone)



Other IoT & Smart
Devices on Network

CURRENT PROJECT: HELM



Home Edge Lifecycle Monitor
(Final Stages)

- NetFlow / IPFIX Telemetry
- Logstash Pipelines & Codecs
- ELK Stack (Dockerized)
- ARM-64 Debian Linux
- IoT Device Monitoring
- Custom Dashboards
- Alerting & Notifications
- Data Filtering & Enrichment
- Telemetry Validation
- Port Mirroring (Testing Phase) ✓

Goal: Full visibility and actionable insights from edge to endpoint.

MONITORING & ANALYTICS STACK



All services containerized with Docker for portability, consistency, and easy management.

FUTURE PROJECTS & ROADMAP



N.O.M.A.D.
(Offline Survival Home Lab)



SPLUNK
ENTERPRISE



PROXMOX
VE



TRUENAS
SERVER



VPN
SERVICES



HONEY POT
ENVIRONMENT



FIREWALL
LAB



CISCO
SWITCH



VLANs &
NETWORK SEGMENTATION



AUTOMATION
& SCRIPTING

LEGEND

- Internet / WAN
- Routing / Gateway
- Wired Network
- PoE / Power + Data
- Wireless Connection
- Future / Expansion

"The best way to predict the future is to build it."
- My Home Lab Journey



CONTINUOUS LEARNING. REAL-WORLD SKILLS. STRONGER EVERY DAY.

Document • Build • Monitor • Detect • Respond • Improve